ASignX Normalized Data Dictionary

Based on Normalized-ERD.uml.txt | Updated 22 October 2026

This dictionary follows Normalized-ERD.uml.txt. It covers 16 application tables containing 87 attributes, plus the Supabase-managed auth_users reference containing its id attribute. The source declares 25 relationships.

The table names, attributes, PostgreSQL types, explicit key markings, and relationship endpoints follow the supplied ERD. Additional fields from the application implementation are not included.

Reading the dictionary

- PK means primary key, FK means foreign key, and UK means explicitly declared unique key. Multiple PK fields in one table form a single composite primary key.
- NS means not specified in the ERD. All column defaults are NS because the source declares no default expressions. No UUID generator, timestamp default, role default, or XP default is assumed.
- Null? is No for a primary key, a starred mandatory attribute, or a foreign key with a mandatory parent relationship. It is Yes for an explicit nullable attribute or optional parent relationship. For other attributes, the SQL nullability is NS.
- VARCHAR(n) limits the value to n characters. NUMERIC(6,5) declares six total digits and five fractional digits. TIMESTAMPTZ is a timestamp with time zone. These types do not by themselves define domain checks or default values.
- The source does not specify ON DELETE or ON UPDATE actions, domain checks, indexes, triggers, or access policies. They are not presented as declared constraints in this dictionary.

Supabase managed users reference

Supabase-managed authentication entity. The ERD declares only its unique user identifier; additional Auth attributes are outside this dictionary.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier supplied by Supabase Auth.

Declared keys and references

- Primary key: (id).

Notes

- Managed authentication is external to the application tables. The ERD does not enumerate email, password, metadata, or other Supabase columns.

1 profiles

Application profile associated with an authentication identity. Stores display name, application role, time zone, and profile timestamps.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK FK	No	NS	Profile identifier shared with the linked authentication identity. References auth_users.id.
display_name	VARCHAR(100)	-	NS	NS	Name displayed for the user in the application.
role	VARCHAR(10)	-	NS	NS	Application role associated with the profile; permitted values are not specified in the ERD.
timezone	VARCHAR(64)	-	NS	NS	Time-zone identifier associated with the user profile.
created_at	TIMESTAMPTZ	-	NS	NS	Record creation timestamp.
updated_at	TIMESTAMPTZ	-	NS	NS	Record modification timestamp.

Declared keys and references

- Primary key: (id).
- Foreign key: id -> auth_users.id.

Notes

- The ERD specifies a mandatory one-to-one relationship between auth_users and profiles. The application must ensure each Auth identity receives its profile; the shared PK/FK prevents multiple profiles for one identity.

2 courses

Learning course definitions and their publication status.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the course record.
title	VARCHAR(150)	-	NS	NS	Title displayed for the course or lesson.
description	TEXT	-	NS	NS	Description of the course or achievement.
is_published	BOOLEAN	-	NS	NS	Publication-status flag for the content.
created_at	TIMESTAMPTZ	-	NS	NS	Record creation timestamp.

Declared keys and references

- Primary key: (id).

3 enrollments

Associates a learner profile with a course and records when the learner enrolled.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the enrollment record.
user_id	UUID	FK	No	NS	Identifies the learner or session owner. References profiles.id.
course_id	UUID	FK	No	NS	Identifies the associated course. References courses.id.
enrolled_at	TIMESTAMPTZ	-	NS	NS	Timestamp of enrollment in the course.

Declared keys and references

- Primary key: (id).
- Foreign key: user_id -> profiles.id.
- Foreign key: course_id -> courses.id.

4 lessons

Ordered lesson material within a course, with a configured XP reward and publication status.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the lesson record.
course_id	UUID	FK	No	NS	Identifies the associated course. References courses.id.
title	VARCHAR(150)	-	NS	NS	Title displayed for the course or lesson.
content	TEXT	-	NS	NS	Instructional content of the lesson.
sort_order	INTEGER	-	NS	NS	Relative display position within the parent record.
xp_reward	INTEGER	-	NS	NS	XP reward configured for the lesson.
is_published	BOOLEAN	-	NS	NS	Publication-status flag for the content.

Declared keys and references

- Primary key: (id).
- Foreign key: course_id -> courses.id.

5 lesson progress

Records a lesson within an enrollment, including its start, completion time, and earned XP.

Column name	Data type	Key	Null?	Default	Description
enrollment_id	UUID	PK FK	No	NS	Identifies the learner course enrollment. References enrollments.id.
lesson_id	UUID	PK FK	No	NS	Identifies the associated lesson. References lessons.id.
started_at	TIMESTAMPTZ	-	NS	NS	Timestamp when the lesson activity or recognition session started.
completed_at	TIMESTAMPTZ	-	NS	NS	Timestamp associated with lesson completion.
xp_earned	INTEGER	-	NS	NS	XP recorded for this enrollment and lesson.

Declared keys and references

- Primary key: (enrollment_id, lesson_id).
- Foreign key: enrollment_id -> enrollments.id.
- Foreign key: lesson_id -> lessons.id.

Notes

- The pair (enrollment_id, lesson_id) is one composite primary key; neither component alone is the primary key.

6 signs

Shared ASL vocabulary, meanings, instructions, categories, and demonstration paths.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the sign record.
gloss	VARCHAR(100)	UK	NS	NS	Unique vocabulary label identifying the sign.
meaning	TEXT	-	NS	NS	Meaning associated with the ASL sign.
category	VARCHAR(60)	-	NS	NS	Category label used to organize signs.
instructions	TEXT	-	NS	NS	Instructions for performing the sign.
demo_path	TEXT	-	NS	NS	Path to a sign demonstration resource.
is_published	BOOLEAN	-	NS	NS	Publication-status flag for the content.
created_at	TIMESTAMPTZ	-	NS	NS	Record creation timestamp.

Declared keys and references

- Primary key: (id).
- Unique key: (gloss).

7 lesson signs

Junction table connecting lessons and signs, with a display order for each linked sign.

Column name	Data type	Key	Null?	Default	Description
lesson_id	UUID	PK FK	No	NS	Identifies the associated lesson. References lessons.id.
sign_id	UUID	PK FK	No	NS	Identifies the associated sign. References signs.id.
sort_order	INTEGER	-	NS	NS	Relative display position within the parent record.

Declared keys and references

- Primary key: (lesson_id, sign_id).
- Foreign key: lesson_id -> lessons.id.
- Foreign key: sign_id -> signs.id.

Notes

- The pair (lesson_id, sign_id) is one composite primary key. This junction resolves the many-to-many association between lessons and signs.

8 exercises

Questions or practice prompts associated with lessons and, optionally, a sign.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the exercise record.
lesson_id	UUID	FK	No	NS	Identifies the associated lesson. References lessons.id.
prompt	TEXT	-	NS	NS	Prompt displayed for the exercise.
sign_id	UUID	FK	Yes	NS	Identifies the associated sign. References signs.id.
sort_order	INTEGER	-	NS	NS	Relative display position within the parent record.

Declared keys and references

- Primary key: (id).
- Foreign key: lesson_id -> lessons.id.
- Foreign key: sign_id -> signs.id.

Notes

- sign_id is explicitly nullable and its parent relationship is optional.

9 exercise options

Answer options belonging to an exercise, including the option correctness flag.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the exercise option record.
exercise_id	UUID	FK	No	NS	Identifies the exercise. References exercises.id.
option_text	TEXT	-	NS	NS	Text displayed for an answer option.
is_correct	BOOLEAN	-	NS	NS	Indicates whether this answer option is correct.
sort_order	INTEGER	-	NS	NS	Relative display position within the parent record.

Declared keys and references

- Primary key: (id).
- Foreign key: exercise_id -> exercises.id.

10 exercise attempts

Learner submissions linking an enrollment, an exercise, and the selected answer option.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the exercise attempt record.
enrollment_id	UUID	FK	No	NS	Identifies the learner course enrollment. References enrollments.id.
exercise_id	UUID	FK	No	NS	Identifies the exercise. References exercises.id.
selected_option_id	UUID	FK	No	NS	Identifies the selected exercise option. References exercise_options.id.
attempted_at	TIMESTAMPTZ	-	NS	NS	Timestamp of the exercise or camera-practice attempt.

Declared keys and references

- Primary key: (id).
- Foreign key: enrollment_id -> enrollments.id.
- Foreign key: exercise_id -> exercises.id.
- Foreign key: selected_option_id -> exercise_options.id.

Notes

- This table has no is_correct attribute in the supplied ERD. The selected option is recorded; its correctness flag belongs to exercise_options.
- The ERD shows separate references to exercises and exercise_options. It does not specify a composite foreign key enforcing that the selected option belongs to the recorded exercise.

11 achievements

Achievement definitions, identified by a unique code, with a metric and target threshold.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the achievement record.
code	VARCHAR(50)	UK	NS	NS	Unique stable code identifying the achievement.
name	VARCHAR(100)	-	NS	NS	Display name of the achievement.
description	TEXT	-	NS	NS	Description of the course or achievement.
metric	VARCHAR(25)	-	NS	NS	Metric used to assess achievement eligibility; allowed values are not specified in the ERD.
threshold	INTEGER	-	NS	NS	Target value associated with the achievement metric.

Declared keys and references

- Primary key: (id).
- Unique key: (code).

12 user achievements

Junction table recording which achievement a learner has unlocked and when it was unlocked.

Column name	Data type	Key	Null?	Default	Description
user_id	UUID	PK FK	No	NS	Identifies the learner or session owner. References profiles.id.
achievement_id	UUID	PK FK	No	NS	Identifies the awarded achievement. References achievements.id.
unlocked_at	TIMESTAMPTZ	-	NS	NS	Timestamp when the user unlocked the achievement.

Declared keys and references

- Primary key: (user_id, achievement_id).
- Foreign key: user_id -> profiles.id.
- Foreign key: achievement_id -> achievements.id.

Notes

- The pair (user_id, achievement_id) is one composite primary key. This junction resolves the many-to-many association between profiles and achievements.

13 model versions

Model release definitions, artifact paths, and recognition confidence thresholds.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the model version record.
version	VARCHAR(50)	UK	NS	NS	Unique label identifying the model version.
artifact_path	TEXT	-	NS	NS	Path to the model artifact.
confidence_threshold	NUMERIC(6,5)	-	NS	NS	Confidence threshold associated with the model version.
created_at	TIMESTAMPTZ	-	NS	NS	Record creation timestamp.

Declared keys and references

- Primary key: (id).
- Unique key: (version).

14 practice attempts

Camera-practice outcomes linking a learner, target sign, optional prediction, and model version. Lesson context is optional.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the practice attempt record.
user_id	UUID	FK	No	NS	Identifies the learner or session owner. References profiles.id.
lesson_id	UUID	FK	Yes	NS	Identifies the associated lesson. References lessons.id.
target_sign_id	UUID	FK	No	NS	Identifies the sign the learner was asked to perform. References signs.id.
predicted_sign_id	UUID	FK	Yes	NS	Identifies the predicted sign, when one is available. References signs.id.
model_version_id	UUID	FK	No	NS	Identifies the model version used for the prediction. References model_versions.id.
confidence	NUMERIC(6,5)	-	NS	NS	Confidence value recorded for a practice attempt or recognition result.
attempted_at	TIMESTAMPTZ	-	NS	NS	Timestamp of the exercise or camera-practice attempt.

Declared keys and references

- Primary key: (id).
- Foreign key: user_id -> profiles.id.
- Foreign key: lesson_id -> lessons.id.
- Foreign key: target_sign_id -> signs.id.
- Foreign key: predicted_sign_id -> signs.id.
- Foreign key: model_version_id -> model_versions.id.

Notes

- This table has no is_correct attribute in the supplied ERD. Correctness can be evaluated from the target, prediction, confidence, and referenced model threshold.
- The optional lesson reference allows a practice attempt without a lesson context. The predicted sign may also be NULL.

15 recognition sessions

Saved recognition sessions associated with a user profile, including session times and final text.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the recognition session record.
user_id	UUID	FK	No	NS	Identifies the learner or session owner. References profiles.id.
started_at	TIMESTAMPTZ	-	NS	NS	Timestamp when the lesson activity or recognition session started.
ended_at	TIMESTAMPTZ	-	NS	NS	Timestamp associated with the end of the recognition session.
final_text	TEXT	-	NS	NS	Final text associated with the recognition session.

Declared keys and references

- Primary key: (id).
- Foreign key: user_id -> profiles.id.

16 recognition results

Ordered recognition results within a session, with an optional recognized sign, model version, confidence, and recognition time.

Column name	Data type	Key	Null?	Default	Description
id	UUID	PK	No	NS	Unique identifier of the recognition result record.
session_id	UUID	FK	No	NS	Identifies the parent recognition session. References recognition_sessions.id.
sign_id	UUID	FK	Yes	NS	Identifies the associated sign. References signs.id.
model_version_id	UUID	FK	No	NS	Identifies the model version used for the prediction. References model_versions.id.
sequence_no	INTEGER	-	NS	NS	Position of the result within the recognition session.
confidence	NUMERIC(6,5)	-	NS	NS	Confidence value recorded for a practice attempt or recognition result.
recognized_at	TIMESTAMPTZ	-	NS	NS	Timestamp of the recognition result.

Declared keys and references

- Primary key: (id).
- Foreign key: session_id -> recognition_sessions.id.
- Foreign key: sign_id -> signs.id.
- Foreign key: model_version_id -> model_versions.id.

Notes

- sign_id is explicitly nullable and its parent relationship is optional. The ERD does not mark (session_id, sequence_no) as unique.

Relationship reference

Parent to child is the cardinality shown in the ERD. Required or optional describes whether a child must reference a parent. Delete and update actions are not declared in the source.

Parent	Child foreign key	Cardinality	Reference	Label
auth_users	profiles.id	1 to 1	Required	has profile
profiles	enrollments.user_id	1 to 0..many	Required	enrolls
courses	enrollments.course_id	1 to 0..many	Required	receives
courses	lessons.course_id	1 to 0..many	Required	contains
enrollments	lesson_progress.enrollment_id	1 to 0..many	Required	tracks
lessons	lesson_progress.lesson_id	1 to 0..many	Required	progress for
lessons	lesson_signs.lesson_id	1 to 0..many	Required	includes
signs	lesson_signs.sign_id	1 to 0..many	Required	appears in
lessons	exercises.lesson_id	1 to 0..many	Required	contains
signs	exercises.sign_id	0..1 to 0..many	Optional	reference sign
exercises	exercise_options.exercise_id	1 to 0..many	Required	offers
enrollments	exercise_attempts.enrollment_id	1 to 0..many	Required	makes
exercises	exercise_attempts.exercise_id	1 to 0..many	Required	attempted in
exercise_options	exercise_attempts.selected_option_id	1 to 0..many	Required	selected option
profiles	user_achievements.user_id	1 to 0..many	Required	earns
achievements	user_achievements.achievement_id	1 to 0..many	Required	awarded as
profiles	practice_attempts.user_id	1 to 0..many	Required	performs
lessons	practice_attempts.lesson_id	0..1 to 0..many	Optional	practiced in
signs	practice_attempts.target_sign_id	1 to 0..many	Required	target sign
signs	practice_attempts.predicted_sign_id	0..1 to 0..many	Optional	predicted sign
model_versions	practice_attempts.model_version_id	1 to 0..many	Required	evaluates with
profiles	recognition_sessions.user_id	1 to 0..many	Required	starts
recognition_sessions	recognition_results.session_id	1 to 0..many	Required	contains
signs	recognition_results.sign_id	0..1 to 0..many	Optional	recognized sign
model_versions	recognition_results.model_version_id	1 to 0..many	Required	generated by

Details requiring implementation decisions

- auth_users is the name used in the supplied ERD for Supabase-managed authentication. Only its id is shown; no additional authentication fields are added here.
- The relation auth_users ||--|| profiles requires one profile per Auth identity and one Auth identity per profile in the conceptual design.
- The explicit unique attributes are signs.gloss, achievements.code, and model_versions.version. No additional composite uniqueness is declared.
- completed_at, ended_at, final_text, and descriptive fields have no explicit nullable annotation in the source. Their SQL nullability remains NS here; lifecycle-based optionality should be decided when implementing the schema.
- Business checks such as option/exercise consistency, lesson/enrollment course consistency, score ranges, quiz publication rules, XP award rules, and consent for saved history require separate implementation decisions.